RealQM Molecule Simulator: Validation on Small Molecules from Hydrogen and the Second Row

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Abstract

We describe a simulator for small molecules built from hydrogen, carbon, nitrogen and oxygen in RealQM, a real-space model in which each electron is a unit-charge density occupying a non-overlapping spatial domain, and the validation it is built to support. The central methodological choice is to stop treating the grid as a convergence parameter and instead *declare* it: one spacing, $h = 0.1\,a_0$ on a 200^3 grid in a box of $20\,a_0$, for every molecule in the list. This is not a computational detail but a commitment about what the model is. At fixed h the grid size no longer carries resolution—it fixes the box—and the box must be fixed too, because mixed boxes were measured to corrupt a cross-molecule energy difference by a factor of 2.7. The declaration costs something and buys something. It costs the claim of being parameter-free: h becomes a constant of the model, fixed by measurement rather than derived, and we say so plainly rather than presenting the grid as a numerical setting. It buys falsifiability, since with h free any disagreement can be deferred to convergence, whereas with h declared every bond length and every binding energy is a prediction carrying no remaining adjustment. On that footing the simulator reports both observables directly: binding energy against the measured well depth D_e, and bond length against R_e. We work hydrogen through in full as the foundational case. Its binding energy is $0.1666 \pm 0.0018\,E_h$, where the uncertainty is measured scatter over a factor of two in the initialisation and not an estimate. That value falls *between* the two experimental conventions—4.5% below the well depth D_e and 1.2% above the directly measured D_0—which differ by more than the model's own scatter, so the model is accurate to a few percent on either reading and the sign of its error is set by a convention it cannot resolve. The declared box is worth a further four points, more than the scatter, and runs the unflattering way: a smaller box binds more strongly, so enlarging it costs accuracy rather than buying it. Its bond is 52% too long; and these are not independent defects but one, since the well's curvature is 49% too soft and $\Delta R = F/k_{\mathrm{model}}$ turns a given force residual into a proportionately larger displacement—a conversion that must use the model's own curvature, the measured force constant understating the bond error by the stiffness ratio. Its bond error is almost entirely discretisation rather than model: an axisymmetric reduction, exact for a two-domain system on an axis, converges the same bond to $1.368\,a_0$—short by 2.3% and flat to 0.18% under a 2.5-fold refinement—with the energy minimum and the force zero agreeing to 0.06%. So the refinement sequence crosses experiment on its way to something shorter, and the spacing at which the error vanishes marks a cancellation rather than an agreement. We retain forces at observed geometries as a per-molecule check and give the symmetry rule that decides when the total-force residual measures anything, but we withdraw it as a figure of merit: an unbound hydrogen molecule scores as well on it as the best entries in the list.

1 The simulator

The simulator presents a list of molecules, each stored as experimental internal coordinates converted to Cartesian form and centred on the centroid. Selecting one builds the system on the declared grid of Section 4—$h = 0.1\,a_0$, 200^3, box $20\,a_0$, identical for every entry—and runs. The solver is a GPU code (`mol_fast`) carrying one domain per entry and capped at sixteen domains; a second implementation of the same model as one globally partitioned field admits two thousand and is discussed in Section 7.

Four choices are built in as defaults rather than offered as options, each because it was measured rather than assumed. The grid is fixed, for the reasons of Section 4, and does not follow the molecule—an earlier version sized the box from each molecule's own extent, which is what makes a table of cross-molecule differences unusable. Valence charges are $Z_x = 1, 4, 3, 3$ for H, C, N, O—oxygen at +3 rather than the common +2, which converges five times worse at the same bond error, and carbon never at +2, which has no defence (Section 13). The nuclear force is the volume integral of Section 3.1 rather than a sampled potential gradient. And the kinetic energy of partially occupied interface cells is excluded above an occupancy of 0.9, which is the single change that brings hydrogen's binding energy from -7.6% to within its own scatter of D_e (Section 6).

The kernels are held fixed by default, since the measured geometry is the input the whole method rests on. They can also be released to move under the force, which is the sharpest available test of the force and is reported for H_2 in Section 14. It is offered with the restriction measured there attached: relaxation is trustworthy for geometry but not for energies, and at three centres the electronic state does not settle, so released kernels are at present a two-centre capability rather than a feature of the molecule list as a whole.

The two observables that can be compared with experiment are reported directly in the output. The bond lengths and angles are displayed against their measured values, and the binding energy is displayed as

$$E_{\mathrm{bind}} \;=\; \sum_i E(\mathrm{atom}_i) \;-\; E(\mathrm{molecule}), \tag{1}$$

positive for a bound molecule, against the measured well depth D_e where a verified value exists. The atomic references in (1) are stored *keyed by grid*, and a run whose grid has no reference set reports no binding energy together with the reason, rather than silently subtracting an energy computed on a different mesh. This is the declaration of Section 4 doing its work: once the grid is fixed, the atomic energies are constants of the model rather than per-run bookkeeping.

What further distinguishes the simulator from a calculator is that it displays its own error checks continuously alongside the result: the kernel–kernel energy against exact arithmetic on the geometry actually built, and the total nuclear force as a percentage of the largest single force. The reasoning behind those two quantities, and why which of them carries information depends on the molecule's symmetry, is the subject of Section 9, where we also explain why the second is useful as a convergence diagnostic but must not be read as a figure of merit. Their practical effect is that a calculation which cannot be trusted says so on the same screen as its answer.

2 Introduction

RealQM [1] replaces the many-electron wavefunction by N unit charge densities occupying non-overlapping spatial domains with free interfaces, interacting by Coulomb's law with each pair counted once. Nuclei enter as kernels; in molecular work a filled core is replaced by a *pseudo-kernel* of reduced charge Z_x carrying only the bonding valence. The model is therefore a force law: kernels

move under the electrostatic force of the charge distribution, and a predicted geometry is wherever that force vanishes.

Validating such a model is awkward by the usual routes. Scanning the energy over geometry requires every point of the scan to be converged, and a scan assembled from partially relaxed states returns a smooth but meaningless curve. Relaxing under the force requires the force to be trustworthy, which is the thing under test. Both routes also need many calculations per molecule, and both return an answer whether or not the underlying states deserve one.

We therefore validate on three quantities, which between them fix the shape of the energy well and are reported for every molecule the simulator runs.

Binding energy—the depth. E_{bind} from (1) against the measured well depth D_e. It requires atomic references computed with the same pseudokernel on the same grid, which is why Section 5 treats the atom model before any molecule.

Relaxed geometry—the position. The separation the model itself settles at when the kernels are released, against R_e. This is the model's own answer and the one that must be reported, in preference to a relaxation started at the experimental geometry, which stops short and flatters the model.

Forces at the observed geometry—the gradient. At the measured equilibrium the true force on every nucleus is zero, so the computed residual is the model's error with experiment supplying the reference. No minimum is sought and no scan is assembled; a single calculation suffices. The residual is also quantitative, through

$$\Delta R \;\approx\; F/k_{\mathrm{model}}, \tag{2}$$

where k_{model} is the *model's own* curvature and not the measured force constant. This distinction is not pedantry: for H_2 the model's well is 49% too soft, so the measured k understates the bond error by a factor of 1.97, and (2) with the model's curvature predicts the model's own minimum to 0.2% (Section 6.3).

The three are not independent. A model can have the right depth and the wrong position if its curvature is wrong, and that is exactly what hydrogen does; reporting any one of them alone invites the reader to generalise from it. Section 6 works all three through for H_2, which is the one system in the list whose atomic reference involves no pseudokernel at all and whose errors are therefore attributable to the discretisation rather than to the core model.

The protocol suits a force-law model particularly well, because it tests the model on its own terms: the force is primary, and agreement with $-dE/dR$ is a consistency check rather than the definition. We return to that distinction in Section 15.

3 Model and method

Each electron m carries a density u_m^2 supported on a domain fixed by a weight w_m, with $\int w_m u_m^2 = n_m$ the domain's occupation. Domains compete for territory, so the partition has free internal boundaries. The energy is

$$E = \sum_m \Big[\tfrac{1}{2}\int w_m |\nabla u_m|^2 - \int w_m K u_m^2\Big] + E_{ee} + \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \qquad K(\mathbf{r}) = \sum_a \frac{Z_a}{r_{\mathrm{soft},a}}, \tag{3}$$

with every term weighted by w, electron–electron repulsion counting each pair once by the $i \neq j$ rule, and $r^2_{\mathrm{soft}} = |\mathbf{r} - \mathbf{R}_a|^2 + (\sigma h)^2$ a kernel softening on the grid of spacing h. Calculations use bare kernels ($r_c = 0$) throughout, the declared grid of Section 4, and relaxation to a fixed step count with early termination disabled—the latter because the built-in stopping criterion compares energies between wall-clock samples and was found to report convergence on a plateau from which the state was still moving.

3.1 How the forces are computed

The force on a nucleus is the classical electrostatic force of the charge distribution, the content of the electrostatic force theorem [4, 5]; in a model built from charge densities it is definitional rather than derived. It is obtained by differentiating the discretised energy analytically rather than by sampling its gradient. From $V_{eK} = -\sum_a Z_a \int \rho / r_{\mathrm{soft},a}$,

$$\mathbf{F}_a \;=\; -\frac{\partial V_{eK}}{\partial \mathbf{R}_a} \;=\; Z_a \int \rho(\mathbf{r})\, \frac{\mathbf{r} - \mathbf{R}_a}{r^3_{\mathrm{soft},a}}\, dV, \tag{4}$$

a volume integral over the grid. It carries no averaging radius and is never evaluated at the nuclear cusp, and because it differentiates the same softened kernel from which the energy is built it agrees with $-dE/dR$ by construction, up to the term arising from the partition boundary moving with the nuclei—the analogue here of the basis-set derivative term of Pulay [6].

The alternative of sampling $Z_a \nabla V_H$ at the nucleus and averaging over a sphere of radius a to suppress grid noise is not a well-defined quantity on this mesh: holding H_2 fixed at $R = 1.68\, a_0$ and varying only a from 0.24 to $0.96\, a_0$ carries the force from -0.050 to $+0.027\, E_h/a_0$, with no plateau and a change of sign, since the averaging bias is $O(a^2 \nabla\rho)$ and $\nabla\rho$ is largest exactly at the cusp. Equation (4) is used throughout; its agreement with the energy is measured in Section 6.3, where it predicts the model's own minimum to 0.2%.

4 One mesh for every molecule

All calculations reported here use the GPU solver `mol_fast` on a single declared grid:

$$h = 0.1\, a_0, \qquad 200^3 \text{ cells}, \qquad \text{box } 20\, a_0, \qquad \sigma = 1 \;\; (\varepsilon = h), \tag{5}$$

the same for every molecule in the list. The grid is not adjusted per system and no mesh ladder is run to justify a number. This requires defending, because it inverts the usual practice.

At fixed h, the grid size is the box. Since box $= nh$, declaring h means that choosing between 100^3, 200^3 and 300^3 is choosing a box of 10, 20 or $30\, a_0$—how much vacuum surrounds the molecule, not how finely it is resolved. The bond length is nearly insensitive to that: holding $h = 0.1$ and enlarging the box from 10 to $14\, a_0$ moves H_2 from 1.8855 to $1.8921\, a_0$, a change of 0.35%. Box 20 is chosen as the smallest that holds the polyatomic entries with room for their electron clouds.

The box must be fixed too. This is the reason the grid cannot follow the molecule. Comparing two systems computed in differently sized boxes inflates their energy difference: CO gave $0.0369\, E_h$ on mixed boxes against $0.0137\, E_h$ on matched ones, a factor of 2.7. A table of binding energies assembled from per-molecule boxes is therefore not a table of model errors but of box errors, and the only remedy is one box for the whole table, sized by its largest member.

What the declaration costs. It costs the description of the model as parameter-free. With h declared, h is a constant of the model, fixed by measurement and not derived from anything; and (5) is not an innocent choice, since $h = 0.1$ is the spacing at which the binding energy of hydrogen agrees with experiment while its bond is 52% too long (Section 6)—and Section 6.2 shows the bond error there to be discretisation, the converged value being short by only a few percent. A declared mesh is therefore not neutral between observables: this one is chosen where the energy is right and the geometry is not. A genuine constant of nature would be pinned by something physical; the natural candidate here is the kernel softening length $\varepsilon = h$, and at $0.1\,a_0$ that exceeds a real nuclear radius by four orders of magnitude, so no physical length currently fixes it.

And h is not the only one. The free boundary is advanced by

$$w \;\leftarrow\; w \;+\; \lambda\,\delta t\,|c|\,\nabla^2 w \;+\; \mu\,\delta t\,c\,|\nabla w|, \qquad c \;=\; u_m - \max_{n\neq m} u_n, \tag{6}$$

clamped to $[0, 1]$, with the occupancy set to zero below a floor w_f wherever another domain clearly owns the cell. The fixed point $c = 0$ is the amplitude-continuity condition of Section 6.5 and is part of the model; the coefficients are not. Declaring the grid therefore means declaring, alongside h: the interface sharpness $\lambda = 0.05$, the front speed $\mu = 1$, the residue floor $w_f = 0$, the relaxation step $\delta t = \nu h^2$, the initial territory extent $R_{\text{init}} = 3\,a_0$—and the box itself, which Section 6.1 measures at four points of hydrogen's binding energy and which therefore cannot be presented as a container sized to fit the molecules. We list them because an unlisted regularisation is indistinguishable from a fitted one.

Their measured sensitivity on H_2 is the reason this is a short list rather than a serious one. Reducing λ by a factor of four moves the bond by 0.4 points; doubling R_{init} moves it by 0.2 points and the energy by $0.006\,E_h$. The floor is the exception: raising it from 0 to 0.05 lengthens the bond by up to 2.9 points, monotonically and saturating, and—awkwardly—the setting that leaves no spurious residue behind a retreating front makes the bond *worse*, which is why $w_f = 0$ is declared rather than defaulted quietly. The front speed μ has not been laddered on H_2 at all; it was fixed on a one-electron test where $\mu = 2$ was stable and $\mu = 10$ failed to conserve charge, and that is the outstanding gap in this list.

What it buys. Falsifiability. With h free, any disagreement with experiment can be deferred to convergence, and no single calculation is ever answerable for its result. With h declared, every bond length and every binding energy in the table is a prediction carrying no remaining adjustment, and the whole table stands or falls together. The sharp test, which we cannot yet run, is whether *one* spacing fits both observables at once; Section 6 shows that hydrogen's two observables currently want different ones.

5 The atom model with a pseudokernel

The binding energy (1) is a difference against separated atoms, so the atomic references are part of the model and are computed by the same solver on the same grid (5), with the same pseudokernel: a kernel of reduced charge Z_x carrying only the bonding valence, with $Z_x = 1, 4, 3, 3$ for H, C, N, O taken from atomic calculations made independently of any molecular input [2]. The filled core is not represented; it is absorbed into the reduction of the kernel charge.

The valence is carried as a *single domain of occupancy Z_x*, not as Z_x one-electron domains. This is worth stating precisely, because the two are different constructions and the difference is not

where one might guess. The solver's intra-domain repulsion for an n-occupancy domain retains $(n-1)/n$ of its self-Hartree, and that factor is *exact* for n electrons sharing one spatial density ρ/n:

$$\sum_{i<j}\iint \frac{\rho_i \rho_j}{r} \;=\; \frac{n(n-1)}{2}\cdot\frac{1}{n^2}\iint\frac{\rho\rho}{r} \;=\; \frac{n-1}{2n}\iint\frac{\rho\rho}{r}. \tag{7}$$

So the blob is not missing its electron–electron repulsion, and splitting the valence into one-electron domains would not supply anything absent. What splitting changes is the *shapes*: n non-overlapping domains repel each other less, being further apart, and cost more kinetic energy, being more confined. The blob construction is therefore exact precisely when the n valence electrons share one shape, and in error to the extent that a real valence shell does not. Whether that is the origin of the over-binding reported below is not established.

Three properties of these references bear directly on every binding energy that uses them.

Hydrogen is not a pseudokernel atom. At $Z_x = 1$ the kernel is the bare proton and the single domain is the whole space, so the hydrogen reference involves no core approximation at all. Its error is pure discretisation: $E(\mathrm{H}) = -0.4865\,E_h$ against the exact -0.5, too shallow by 2.7%. This makes H_2 the one system in the list whose binding energy is free of core-model error, which is why we work it through first.

The multi-electron references are strongly over-bound, and the cause is the bare kernel. Measured against the sum of their experimental ionisation energies, carbon on the declared grid is over-bound by a factor of 2.78 and nitrogen by about 2.0, while hydrogen is right to a few percent. It is *not* a missing intra-atomic repulsion, by the identity above. The cause is that the valence *collapses into the vacant core*. From $V_{eK} = -Z_x\sum_i\langle 1/r_i\rangle$, carbon's measured $V_{eK} = -35.33\,E_h$ gives $\langle r\rangle = 0.45\,a_0$ against a real $2s/2p$ mean radius near 1.6: the valence sits some three and a half times too close in, and at that radius a bare +4 kernel would give -35.56, so the collapse accounts for essentially the whole error. In a real atom orthogonality to the $1s$ pair excludes the valence from the core region. The pseudokernel has no core and, with $r_c = 0$, nothing performs that exclusion.

Restoring it confirms the diagnosis. With the inner cutoff r_c reinstated and the rest of the construction untouched:

r_c (a_0)	E(C)	vs reference	error	$\langle r\rangle$
0	−15.106	2.778	+178%	0.45
0.7	−6.702	1.233	+23%	1.15
0.9	**−5.506**	**1.013**	**+1.3%**	1.43
1.1	−4.702	0.865	−13.5%	1.71

with the reference $5.437\,E_h$ and the crossing bracketed at $r_c = 0.91$–0.92 from either adjacent pair. So a single-domain valence reproduces carbon's binding to 1.3% with one parameter, and the statement that "bare kernels" is a harmless simplification is wrong: it is the largest single error in this work, and it propagates to the binding energies of Section 11. Hard depletion of the weight inside r_c outperforms clamping the kernel by two to one at the same radius (+23% against +48%), because the clamp merely caps the attraction at $-Z_x/r_c$ and lets density remain inside.

But the radius does not transfer to molecules, and it is not a core radius. Carbon's $1s$ mean radius is $0.263\,a_0$ on a Slater estimate, so the calibrated 0.92 is 3.5 times the core it is supposed to represent—and oxygen's is four times. At that size the exclusion spheres on two heavy atoms nearly meet: N_2's bond is $2.075\,a_0$ against $2r_c = 1.574$, leaving $0.50\,a_0$ of bonding region, and the

calculation then converges *reproducibly to an asymmetric state* ($\Sigma F = 88\%$ of the largest force on a homonuclear molecule, twice, where symmetry forces it to zero). X–H bonds are unaffected because hydrogen has no core and carries $r_c = 0$, so the one-sided exclusion leaves 1.0–1.15 a_0; it is heavy–heavy bonds that fail. At the *physical* radius the recovery is only about a third of the error—the rest was being bought by making the radius too large—so a single parameter cannot both repair the atom and survive in a molecule.

Nor is the radius established as physical. It was fixed by matching an energy, and this grid has a resolution error of its own: helium, whose construction involves no core at all, comes out 10.5% too shallow here with its kinetic energy 24% low, because $\varepsilon = h = 0.1$ cannot resolve a $Z = 2$ cusp—hydrogen is 2.7% shallow at the same softening. A $Z_x = 4$ cusp is sharper still, so $r_c = 0.92$ is absorbing that error along with the core exclusion, and it must be read as this grid's value until a mesh ladder on the calibration says otherwise. What does not depend on the grid is the mechanism: an error of 178% is far too large to be a softening artefact, and the collapse is visible directly in $\langle r \rangle$.

The same defect is present in the molecule, so it cancels substantially in the difference (1); but that cancellation is precisely what a binding energy *measures*, and it must not be assumed. This is why differences taken within one composition are trustworthy while differences across compositions are not.

Nitrogen and oxygen are degenerate. Both carry $Z_x = 3$ with three one-electron domains, so their atomic calculations are bit-identical. The model as it stands cannot distinguish nitrogen from oxygen at the atomic level, and no quantity computed from these two references speaks about that difference. Oxygen's shell structure is 2+3+3 in the atomic work [2]; reducing it to a +3 valence kernel discards exactly the information that separates the two elements.

6 Hydrogen

We work hydrogen through on the three observables the simulator reports: the binding energy, the geometry the model relaxes to, and the force residual at the observed geometry. H_2 is the natural first case because, as Section 5 notes, its atomic reference carries no pseudokernel, so every error below belongs to the discretisation and the interface treatment rather than to the core model.

Everything below is on the declared grid (5). Before the results, what the box contributes at fixed spacing, because it turns out to contribute more than a grid choice should:

box (a_0)	E(H)	$E(H_2)$	E_{bind}	vs D_e	R
10	−0.486810	−1.147250	0.173630	−0.5%	1.7951
12	−0.486509	−1.145757	0.172739	−1.0%	1.7963
14	−0.486591	−1.143770	0.170588	−2.2%	1.7973
20 (declared)	**−0.486591**	**−1.139790**	**0.166608**	**−4.5%**	**1.7983**

The atomic reference is box-free, varying by $0.0003\,E_h$ across the range, and the bond is box-free, varying by 0.18%. The whole effect is in the molecular energy, which is the interior interface again: the molecule has one and the atom does not. The binding energy spans 4.0 points of D_e, twice the initialisation scatter measured below, so the dependence is real.

It cannot be extrapolated away. The shift per unit box is 0.00075, 0.00099, $0.00066\,E_h$ across the three intervals—not monotone, so not a power law in 1/box—and every individual rung is at or below the scatter even though the overall range is twice it. Neither a convergent limit nor an unbounded expansion of the free boundary is established, and we quote no box-limit value.

Two consequences follow, and both are more awkward than the grid declaration was meant to be. The box joins h and the five coefficients of (6) in the declared set: it is not a convenience sized to hold the molecules but a parameter worth four points of the answer. And the dependence runs the unflattering way—the *smallest* box binds most strongly, within half a percent of D_e, and enlarging it to the declared value costs four points. An earlier figure of six points between boxes 12 and 15 was obtained at fixed *cell count*, so the spacing changed with the box; it is confounded and withdrawn.

The experimental targets are $R_e = 1.40028\,a_0$ and $k = 5.75\,\mathrm{mdyn/\AA} = 0.3692\,E_h/a_0^2$. For the dissociation energy there are two standard values and the difference between them exceeds the model's own scatter, so we give both:

		E_h	eV	
D_0	measured	0.16457	4.478	the directly observed value, from the vibrational ground state
D_e	well depth	0.17449	4.748	D_0 plus a zero-point energy of $0.00993\,E_h$

D_e is the matched comparison, because the model carries no nuclear motion: what it computes is the depth of a well, not the energy of a vibrational state. But D_e is a hybrid of a measurement and a spectroscopic correction, whereas D_0 is purely measured, and the two differ by 6.0 percentage points. We quote against D_e and state the D_0 figure alongside it throughout rather than choose silently.

6.1 Binding energy

interface kinetic treatment	$E(\mathrm{H})$	$E(\mathrm{H_2})$	E_{bind}	vs D_e	vs D_0
cut-off at $w = 0.9$, box 20	**−0.486591**	**−1.139790**	**0.166608**	**−4.5%**	**+1.2%**
cut-off at $w = 0.9$, box 12	−0.486509	−1.145757	0.172739	−1.0%	+5.0%
full w-weighting, box 12	−0.486422	−1.134067	0.161223	−7.6%	−2.0%
exterior cleared, box 12	−0.485361	−0.935158	−0.035564	unbound	unbound

Table 1: Binding energy of $\mathrm{H_2}$ under three treatments of the partially occupied cells at the interface, with molecule and atom relaxed for the same number of steps. Note that the two conventions disagree about which of the first two rows is better, the cut-off being 1.0% low against D_e and 5.0% high against D_0 while the uncorrected weighting is 7.6% low and 2.0% low respectively—another reason to quote both. The reference and the molecule must be run to the same step count: an earlier value of −11.4% for the middle row came from an atomic reference stopped at 70 500 steps against a molecule at 232 500, and is withdrawn.

Table 1 shows that the binding energy is decided by how the interface cells are counted. Where two domains meet, the partition weight w passes through intermediate values, and the kinetic energy of those cells is an artefact of the diffuse representation of a boundary that the model treats as sharp. Excluding cells below an occupancy of 0.9 removes $0.0117\,E_h$ from the molecule and only $0.000087\,E_h$ from the atom—a ratio of 67 to 1 per domain—because the molecule has an interior interface where two domains meet while the atom has only an outer edge with nothing on the far side. The exclusion therefore does not cancel in the difference, and it is the single change that brings the binding energy to experiment.

The third row is the control. Clearing the exterior amplitude altogether, rather than merely declining to count its kinetic energy, unbinds the molecule: E_{bind} goes negative and the relaxed

separation runs out to $3.56\,a_0$. The amplitude just outside a domain is what makes the interface flux small, and removing it imposes a drain at the boundary instead.

The uncertainty is measured, not estimated. The exterior amplitude in `mol_fast` is not varied by the relaxation—where w vanishes the update is identically zero—so its value is inherited from the initial state, and the binding energy could in principle be an artefact of that choice. Varying the initial decay constant by a factor of 2.1 gives

z_{init}	0.7	1.0	1.5
E_{bind}	0.175248	0.172739	0.176339
vs D_e	+0.45%	−0.99%	+1.07%
R (relaxed)	1.7993	1.7963	1.7953

a spread of 2.1 percentage points which is not monotone in the seed, so it is scatter rather than a dependence, and a geometry spread of 0.22%. We therefore quote

$$E_{\mathrm{bind}}(\mathrm{H_2}) \;=\; 0.1666{\pm}0.0018\,E_h \;=\; -4.5{\pm}1.0\%\ \text{of}\ D_e \;=\; +1.2{\pm}1.1\%\ \text{of}\ D_0 \qquad \text{[declared box; +4.0 points at box} \tag{8}$$

the central value on the declared grid, the stated uncertainty the measured initialisation scatter, and the bracketed figure the declared-box sensitivity—which is twice the scatter and must be quoted with it. The result lies *between* the two experimental conventions, $D_0 = 0.16456 < 0.16661 < D_e = 0.17447$, closer to the directly measured D_0 than to the well depth it should properly be compared with. So the model's binding is accurate to a few percent on either reading, and which few percent depends on two things it cannot itself resolve: a convention, and a declared box. For scale, the choice of interface condition discussed in Section 6.5 moves the same quantity by 29 to 58 points, so the initialisation is a minor term.

6.2 Relaxed geometry

Releasing the kernels under the force and letting the system find its own minimum gives the geometry the model actually predicts. For $\mathrm{H_2}$ at the declared grid this is

$$R_{\min} = 2.134\,a_0 \qquad (+52.4\%\ \text{against}\ R_e = 1.40028), \tag{9}$$

taken from a fixed-geometry scan on commensurate separations $D = (2i+1)h$, which avoids the spurious corrugation that appears when kernels are placed at arbitrary sub-cell offsets. The force zero of the same scan lies at $2.066\,a_0$, agreeing with the energy minimum to 3.2%.

A one-sided relaxation released from the observed geometry stops at $1.796\,a_0$ (+28.3%), well short of the minimum, and that is a systematic bias rather than a better answer: released from inside, the molecule expands and the relaxation terminates while the force is still outward. The two-sided bracket is 1.85 to $2.09\,a_0$. We report +52%, which is the model's own answer, and note that any relaxation started at the experimental geometry flatters the model by roughly ten points.

Refining the mesh does not converge on experiment. The one-sided relaxed bond over three spacings is

h	0.1395	0.1000	0.0800
R	1.8897	1.7963	1.6999
dR/dh	—	2.365	4.820

with the local slope *steepening* rather than flattening. Extrapolating linearly in h from the two finest points gives $R(h \to 0) = 1.314\,a_0$, which is 6% *below* experiment, and crosses the measured 1.40028 at $h \approx 0.018$.

That crossing is not convergence on the right answer, and we can now say so from a converged calculation rather than an extrapolation. In three dimensions $h = 0.02$ is out of reach—500^3 cells at the smallest legitimate box—but H_2 is exactly a two-domain system on an axis, which an axisymmetric reduction treats exactly. Scanning commensurate separations $D = (2m + 1)h$ there gives

h	0.050	0.040	0.030	0.025	0.020
$D_{\min}$	1.3707	1.3682	1.3686	1.3687	1.3682
force zero	1.3699	1.3702	1.3691	1.3684	1.3685

a bond flat to 0.18% across a 2.5-fold refinement and already converged by $h = 0.04$. The minimum and the force zero agree to 0.06% at every spacing, which is the sharpest confirmation available that the force is the gradient of the energy. The converged bond is $1.368\,a_0$**, or** -2.3%: short, and not for want of resolution.

So both interface conditions undershoot—-2.3% with the strict condition, -6% extrapolated with the weak one—and neither reaches experiment at any spacing. The $+52\%$ of (9) is therefore almost entirely *discretisation*, and the construction's own bond error is a few percent. The crossing at $h \approx 0.018$ is a positive discretisation error passing through zero as it cancels a negative construction error; declaring that spacing would fit one number to one observable and predict nothing. The caveat belongs on the axisymmetric figure too: it carries a different softening coefficient, but $\varepsilon = \sigma h$ vanishes with h either way, so both schemes approach the same bare-Coulomb problem and the interface condition is the only difference that survives the limit.

6.3 Forces at the observed geometry

The third observable needs no reference calculation and no minimum-finding. At the measured geometry the true force on every nucleus is zero, so the computed residual is model error read directly, with experiment supplying the reference. For H_2 at R_e,

$$F(R_e) = +0.1383\,E_h/a_0 \qquad \text{(outward: the bond wants to lengthen)}, \tag{10}$$

one calculation rather than a scan. The residual converts to a displacement through $\Delta R \approx F/k$, and *which* k appears there matters:

$$\frac{F}{k_{\text{model}}} = \frac{0.1383}{0.1875} = 0.738\,a_0 \quad \Longrightarrow \quad R = 2.138\,a_0, \tag{11}$$

against the scanned minimum of 2.134—agreement to 0.17%. The force at the observed geometry predicts the model's own equilibrium, which establishes that the volume integral of Section 3.1 is the gradient of the energy the solver reports, and it does so from a single calculation.

A correction to the conversion. Using the *measured* force constant instead gives $F/k = 0.375\,a_0$, or $+26.8\%$, which understates the model's bond error by the stiffness ratio $k/k_{\text{model}} = 1.97$. The measured k answers a different question—what bond error this force residual would produce in a well of the correct stiffness—and it is the model's own curvature that governs where the model's minimum lies. Bond errors converted with the measured k are therefore systematically too small whenever the model's well is too soft, and we quote ΔR from the model's own curvature below, at the cost of a three-point scan per molecule in place of one calculation.

6.4 One defect, not three

The three observables are not three independent verdicts. Collecting them against the measured curvature $k = 0.3692\,E_h/a_0^2$:

depth	$E_{\mathrm{bind}} = 0.1666 \pm 0.0018$	$-4.5 \pm 1.0\%$
curvature	$k_{\mathrm{model}} = 0.1875$	-49%
position	$R_{\mathrm{min}} = 2.134$	$+52\%$

the depth is right, the gradient is consistent with the energy to 0.2%, and the position is wrong *because* the well is too soft: a curvature 49% low turns a given force residual into a displacement nearly twice as large. The error is one quantity, the curvature, with the bond length as its consequence.

One qualification, from the converged calculation of Section 6.2: the soft well is a property of the *weak interface treatment*, not of the model. The same system under the strict condition gives curvatures scattering about $0.41\,E_h/a_0^2$, some 11% *above* experiment rather than 49% below. So "one defect, the curvature" is a statement about this solver's front, and the two interface conditions bracket the true curvature from opposite sides exactly as they bracket the bond. The reason the position can be so wrong while the depth is right is that the well is flat: sitting 52% long costs $\frac{1}{2}k\Delta R^2 = 0.051\,E_h$ on the model's own curve, and any quantity differenced near the bottom of a well that shallow is insensitive to where along it the minimum sits.

6.5 The interface condition, and why hydrogen cannot decide it

The free boundary between two domains needs a condition, and the construction admits two that are both defensible. The model requires $\Psi \in H^1$, that is continuity of the densities across the interface, whose natural condition is continuity of the amplitude, $u_1 = u_2$, with the normal derivative free to jump. `mol_fast` implements this: its front advances on the local amplitude mismatch and converges where that mismatch vanishes. The alternative treats the domains as disjoint with u unconstrained at its own boundary, for which the natural condition is $\partial u/\partial n = 0$; a mirrored ghost value, or equivalently omitting the face from the stencil, imposes it exactly, and two independent implementations in this programme do so.

The two give materially different energies—+27.9% and +58.0% on E_{bind} for the strict condition against +0.2% for the continuity condition, with bond errors of +15.9% and −2.3% against +52%—so the choice is not a detail. But H_2 cannot adjudicate it. A symmetric diatomic has $u_1 = u_2$ at the midplane *and*, by reflection, $\partial u/\partial n = 0$; the two conditions coincide, and the spread above is a difference between discretisations rather than between physical conditions. Settling it requires a two-domain system with no symmetry plane, which is not in the present list.

7 The solver

The measurements above were made with a GPU solver (`mol_fast`) that carries one domain per entry and caps at sixteen domains. That cap is the binding constraint on molecular size, and its meaning depends on the construction: with one domain per atom carrying Z_x electrons it permits sixteen atoms, which covers benzene, glycine, ethanol and comparable small organics; with the one-electron domains of the underlying construction it permits sixteen electrons, which covers CH_4 and C_2H_2 but not benzene.

The scope of the pseudo-kernel core model itself, measured on two-centre systems, is reported separately [3].

Two properties of `mol_fast` are relevant to reading the numbers above, and we state them rather than leave them implicit. The partition weights are diffuse: w passes through intermediate values over a few cells at an interface instead of switching between 0 and 1, which is what makes the interface kinetic energy of Section 6.1 a quantity that must be decided rather than a consequence of the model. And the amplitude outside a domain is not relaxed—where w vanishes the update is identically zero—so it is inherited from the initial state. The measured consequence of that is the $\pm 1.0\%$ band on the binding energy, which is small, and bounds how much of the result can be attributed to initialisation rather than to the model.

A second implementation (`molecule.js`) carries the same model as one global field partitioned by labels, enforcing true non-overlap with sharp interfaces, and admits two thousand domains; it has been run on alpha-helices and on chignolin. It is not used for any number reported here. Its interest is that its sharp partition removes the interface ambiguity above at the cost of a different force treatment, so it is the natural setting for the asymmetric two-domain test that Section 6.5 identifies as outstanding.

8 Cost

A volume integral over the grid is more expensive than a point gradient, and it is worth stating by how much. The point gradient visits $N_{\text{atoms}} \times \frac{4}{3}\pi R^3 \approx 270\,N$ cells for a four-cell sphere. The volume form visits $N_{\text{atoms}} \times S^3$, about $10^6 N$ on a 101^3 grid: some four thousand times more.

The relevant comparison, however, is with what the solver already does. One volume-force evaluation costs about the same as one update sweep of the electronic field, and *less* than one evaluation of the potential, which requires an elliptic solve of several sweeps per electron. Forces are moreover not evaluated every step: once per animation frame in one implementation, every ten electronic steps in the other, giving overheads of well under one percent and of order ten percent of the update cost respectively—the latter adjustable, since nuclei move slowly compared with the electronic relaxation.

At protein scale the $N \times S^3$ product would warrant a cutoff radius. It is worth noting that such a cutoff behaves oppositely to the smoothing radius it replaces: truncating a convergent integral has an error that *falls* with radius, and distant neutral groups contribute as dipole/r^3 rather than charge/r^2, so a plateau in the cutoff is attainable and meaningful. The smoothing radius of the point gradient admitted no plateau because its bias *grew* as a^2. The same word denotes a convergent truncation in one case and a divergent bias in the other.

9 Three checks, and which one applies

Every calculation carries three internal checks. They are not interchangeable, and the choice between them is dictated by symmetry—a point we found easy to get wrong.

Kernel–kernel energy. $V_{KK} = \sum_{a<b} Z_a Z_b / R_{ab}$ is exact arithmetic and confirms the geometry *actually built*, independently of any physics. Every configuration reported below matched its predicted value; this is how fractional kernel placement, per-bond spacing and coincident kernels were verified.

Total force as a convergence criterion. The sum of nuclear forces is *not* automatically zero. It equals the net force the electron cloud exerts on the nuclei; only when the electronic state is stationary does the cloud exert no net force, and then $\sum_a \mathbf{F}_a = 0$ follows by Newton's third

law. The residual is therefore a direct measure of electronic non-convergence—and a more reliable one than any energy-based criterion, since a relaxation can sit on a long plateau with the energy changing imperceptibly while the state is far from settled.

But symmetry cancels it. In a symmetric molecule, mirror or rotational symmetry forces $\sum_a \mathbf{F}_a = 0$ regardless of accuracy. A small residual in CO_2 (we measure 0.06% of the largest force) establishes only that the algorithm preserves symmetry. The sum carries information only in directions symmetry does not protect: every direction for a heteronuclear molecule, the C_3 axis of NH_3, the C_2 axis of water.

Per-atom zeros. Where symmetry is complete, the useful check is instead a single atom's force. An atom at a centre of symmetry—the carbon of CO_2 or of CH_4—must feel exactly nothing. Unlike the sum, this cannot be satisfied by cancellation among atoms, so a nonzero value is unambiguously error and its magnitude is the per-atom floor.

10 Results

Table 2 collects the configurations. Forces are quoted with + meaning outward, so a positive entry indicates the bond wants to lengthen. Where a configuration fails its own check we report the failure rather than a bond error.

Two columns of this table require the qualifications established on hydrogen, and we give them before the table rather than after it. The ΔR entries were converted with the *measured* force constant, so by Section 6.3 each is a *lower bound* on the model's bond error, too small by the ratio of the measured curvature to the model's own; for H_2 that ratio is 1.97 and the entry below should read +52% rather than +30%. Correcting the remaining rows requires a three-point scan per configuration to measure each model curvature, which has not been done. And the last column is a diagnostic, not a score: an H_2 configuration that does not bind at all returns 2.1% on it, which would place it among the best entries here. Its only legitimate reading is the symmetry-dependent one given in the caption.

Only the declared valences appear. Every row is at $Z_x = 1, 4, 3, 3$. The C+2 and O+2 variants are not results of the model but the alternatives it rejects, measured as such in Sections 12 and 13, and carrying them here would present a ruled-out convention as a finding. Removing them has one consequence worth stating plainly: **CO and CO_2 leave the table altogether**, because the only force calculations ever made for them were at O+2. There is no force measurement for either molecule at the valences the simulator actually uses, and the earlier reading of CO_2's -7.2% as an ordinary bounded error was therefore a statement about a configuration this paper rejects. What is measured for CO_2 at $3, 4, 3$ is its binding energy, and it does not bind (Section 11).

The binding-energy column carries a third and sharper qualification. Only hydrogen is on the declared grid; the rest predate it, and the single row available on both grids shifts by 35 points of D_e between them. That shift is larger than every deficit in the column, so no entry here establishes that a molecule fails to bind—a statement we had drawn from this set and now withdraw. What the column does establish is the direction and rough scale of the construction's binding on a common grid, and that the earlier grid understated it severely.

10.1 H_4 against an exact reference

One configuration can be scored against an exact method rather than experiment. Full configuration interaction in a minimal basis [9] places the equilibrium of the equally spaced H_4 chain at $a = 1.8\,a_0$, and the chain has one electron per site with no pseudopotential, so no core parameter is involved. At that geometry all four RealQM forces point outward—the model would expand a chain the exact method calls stationary—while over-binding by 14–18% per atom when each method is referred to its own atom. Too long is the pattern H_2 shows at +52% (Section 6.2). The over-binding is less clear-cut than it reads: the reference is a full configuration-interaction calculation in a minimal STO-6G basis, whose cohesive energy is itself 14% low against reality (0.0766 against $0.0872\,E_h$ per atom), so over-binding against it is not the same statement as over-binding against experiment.

11 Binding energies across compositions

Hydrogen's binding energy is accurate because of a cancellation, not because its energies are accurate: the total is 2.4% too shallow and 94% of that error cancels against the atomic reference, leaving the $0.0017\,E_h$ residual that is the binding error. The cancellation is that complete because molecule and reference share a composition and, for H_2, because the reference is a one-electron domain with no pseudokernel at all. Neither holds for the rest of the list, and the arithmetic of how badly it fails is worth setting out, because it decides which binding energies can be believed.

The pseudokernel atoms are over-bound by the factors of Section 5: against the sum of their ionisation energies, carbon by 2.57 and the $Z_x = 3$ oxygen by 2.03. Converting those to absolute terms, each carbon reference is too deep by $8.55\,E_h$ and each oxygen by $3.47\,E_h$, and the same error enters the molecule.

Provisional figures. The two factors were measured on an earlier grid (box 15, $h = 0.1389$, $\varepsilon = 0.14$). Re-run on the declared grid (5) and converged, carbon is $-15.106\,E_h$ rather than -13.999, a factor of 2.78 and an absolute error of $9.67\,E_h$. The finer grid is *deeper*, so the over-binding is worse where this paper's calculations actually live, and the numbers in the table below are therefore lower bounds. The oxygen repeat is in progress. The conclusion drawn from the table is unaffected in direction—a deeper reference makes the required cancellation more stringent, not less—but no figure here should be quoted until both are on one grid.

What matters is the error's size relative to the bond energy being extracted:

system	common error carried in	E_{bind}	ratio
H_2	$0.03\,E_h$	+0.1666	**0.2**
CH_4	$8.61\,E_h$	+0.4081	21
CO_2	$15.50\,E_h$	−0.2110	**73**

The N_2 row of this argument has since been withdrawn. Re-measured with molecule and reference on one grid, N_2 binds by $+0.2764\,E_h$, so the 450× ratio and the 99.8% cancellation requirement computed for it were artefacts of differencing across grids—and its denominator was the model's own near-zero binding, which made the ratio circular as well. What survives is the *structure* of the argument, not its worst case: the common atomic error does grow with composition, and CO_2 at $-0.211\,E_h$ is a deficit an order of magnitude larger than N_2's was, so it is the row that still requires a matched-grid repeat before anything is concluded from it.

For H_2 the common error is a sixth of the signal, so *no* cancellation is needed to get the sign right—the molecule would bind even if the two errors did not cancel at all—and the 94% that does cancel is what turns a sound sign into a 1% magnitude. For CO_2 the common error is seventy-three

times the signal, so the cancellation must be complete to 98.6% merely to return the right *sign*. It is not, and CO_2 comes out unbound by $0.211\,E_h$. Nothing is specifically wrong with CO_2. It is the composition in this list where the atom model's over-binding is largest relative to what the binding energy is trying to measure.

This gives the existing rule—differences within one composition are trustworthy, differences across compositions are not—a quantitative form, and it says where the remedy has to be applied. No improvement to the interface treatment or the grid can rescue CO_2's binding energy, because the defect is in the atomic reference and enters at fifteen Hartree. The pseudokernel atom model is the thing that has to improve—but Section 5 does *not* identify what in it is wrong, and we are careful not to imply otherwise. The two candidates there are testable and untested.

12 Kernel depth decides convergence

The clearest results hold geometry, symmetry, mesh and every other input fixed and change only the central kernel's charge:

	shallow	deep	factor
CH_4 (identical T_d geometry)	C+2: 40%	C+4: 0.6%	60
HCN (identical geometry)	C+2: 88.7%	C+4: 5.5%	16

A deep kernel holds its domain and fixes its boundary against the neighbours; a shallow one leaves the partition undetermined. The direction is worth emphasising: C+4 is the *less* resolved kernel, at $Zh = 0.48$ against a working guideline of $Zh \lesssim 0.1$, so depth helps convergence more than coarse resolution hurts it.

Three alternative explanations were tested and excluded.

- **Not hydrogen content.** CH_4 carries four hydrogens and is the cleanest polyatomic measured; HCN carries one and reaches 5.5%; CO at 4,2 carries none and fails at 38.5%. Hydrogen is neither necessary nor sufficient for failure.
- **Not the box.** The net force on water is 0.216, 0.215, 0.214 in boxes of 12, 18 and $24\,a_0$ with the mesh held fixed—0.9% across a doubling of the box. The state is box-converged and the residual is a property of it. The control calculation on H_2 changes *sign* with box and with mesh, marking its own error as stochastic.
- **Not off-axis placement.** Tetrahedral ligands are maximally off-axis and CH_4 is nearly exact, because the cube diagonals are a symmetry of the grid.

12.1 Cubic-grid anisotropy

A second, smaller effect is visible in planar NH_3, which has D_{3h} symmetry and must therefore give $\sum_a \mathbf{F}_a = 0$ in every direction; it gives 70% of the largest force along x. Resolving by ligand: the one lying on the grid axis has radial force 0.0802, the two off-axis ones 0.0545 and 0.0544. The off-axis pair agrees with itself to 0.2% and differs from the on-axis ligand by 47%.

A cubic grid has fourfold symmetry in its planes and threefold symmetry about its body diagonals. Molecular geometries that share those symmetries—linear, and tetrahedral—are represented evenly; threefold-in-plane and bent geometries are not. This accounts for NH_3 and H_2O being the two worst configurations measured and CH_4 among the best, and it is a property of the discretisation rather than of the model.

13 Valence conventions, measured

Z_x is a modelling choice, and the protocol makes it testable. A depth ladder at fixed mesh, scored both by the symmetry-breaking error and by the bond error:

	O+2	O+3	O+6 (single domain)	N+3	N+5
pair asymmetry	19.5%	**3.7**%	1.4%	6.0%	6.1%
ΔR	−6.4%	**−7.1**%	+59%	−8.1%	−13.7%

O+3 is the better oxygen: a fivefold gain in convergence over O+2 at the same bond error. N+3 stands, full valence costing accuracy while buying no convergence. Most instructive is O+6 as a single six-electron domain, which converges best of all and yields a bond 59% too long—it converges accurately to the wrong answer, which is exactly the failure mode the protocol is designed to make visible. So a rule of "use the deepest Z_x" is false beyond the valence structure the atomic model selects; depth helps only where the domain was genuinely under-bound.

It is worth noting that O+3 is one of the two valence shells in the atomic configuration 2+3+3 obtained independently from atomic calculations [2], a convergence of atomic and molecular evidence on the same number. Carbon admits no corresponding reduction: C+2 would freeze into the core the $2s$ pair that hybridises into the bonds, cannot form four bonds, and—since Z_x is the only label distinguishing elements—makes carbon identical to O+2.

14 Kernel dynamics

Releasing the kernels and letting them move under the force is both the sharpest test of the force and the capability a simulator is finally for. It is also where the distinction between a force and a sampled gradient stops being academic: a force that is not the gradient of anything conserves no energy, so in sustained dynamics it pumps or drains the system indefinitely.

Released from either side of the expected minimum, H_2 relaxes to

start	relaxes to	residual force	direction
1.40	1.8502	$[-0.0178, +0.0057]$	expanded
2.40	2.0932	$[+0.0128, -0.0168]$	contracted

Both move the correct way and bracket $[1.850, 2.093]$, whose midpoint 1.972 sits 0.5% from the energy minimum at 1.963 and inside the force's own zero crossing of 1.84–1.89 measured at fixed geometry. Three independent routes—force zero, energy minimum, and dynamical relaxation from both directions—therefore agree to a few percent, where before the force was repaired the first two disagreed by 27% and the force pointed the wrong way at $R = 1.68$.

The bracket is not a failure to converge but a measured precision. The residual energy drift between runs is $\sim 0.011\,E_h$; through the curvature $k = 0.370\,E_h/a_0^2$ that corresponds to a displacement $\sqrt{2(0.011)/0.370} = 0.24\,a_0$, and the bracket is $0.243\,a_0$ wide. The two relaxations converged to exactly the precision the unconverged electronic state permits, and tightening the geometry requires tightening the electronics, not the dynamics.

Two cautions follow, both measured rather than anticipated. The relaxed states carry energies about $0.011\,E_h$ *above* the fixed-geometry scan at comparable separation, the electronic field lagging behind moving nuclei in the familiar way; so relaxation yields trustworthy geometry and untrustworthy energies, and the two should be taken from different calculations. And at three centres the

electronic relaxation does not settle at all, oscillating by $\sim 0.03\,E_h$ out to 3×10^5 steps, so kernel dynamics there would track a wandering density. Dynamics is at present a two-centre capability.

14.1 What the old force does in dynamics

The same relaxation under the sphere-averaged gradient settles at 1.5203 with residual forces $[+0.01821, +0.00020]$—both of the same sign, so the molecule carries a net translational force and drifts bodily, which no isolated system may do. In sustained dynamics that is a spurious acceleration of the whole system, and in a thermostatted simulation it would appear as heating indistinguishable from physical dissipation.

It also lands *closer to experiment*: 1.520 against the measured 1.401, where the repaired force gives 1.97. This is worth stating plainly because it inverts the naive test. The repaired force agrees with the model's own energy minimum to 0.5%; it reports faithfully what the model says, and what the model says is wrong by 40%. The old force was wrong *about the model* in a way that partly compensated for the model's own error. Validating a force against experiment alone would have preferred the defective one; only the internal checks—the lone atom feeling exactly zero, the vanishing of $\sum_a \mathbf{F}_a$, and agreement with $-dE/dR$—reject it. Fixing the instrument made agreement with measurement worse and moved the error from the instrument to the model, which is where it can be addressed.

15 Relation to $-dE/dR$

In a force-law model the force is definitional and the geometry is where it vanishes, so agreement with the energy gradient is not required for the model to be well posed. It remains worth checking for three reasons. A force that is not a gradient conserves no energy, which in dynamics produces spurious heating indistinguishable from physical dissipation. It is a strong and free check on implementation: it is what exposed the sign-flipping force of Section 3.1, which was otherwise unremarkable—smooth, antisymmetric, and crossing zero at a plausible separation. And once the implementation is correct, the residual disagreement measures the contribution from the partition boundary moving with the nuclei, a term specific to a free-boundary model and not otherwise accessible.

16 Bearing on previously recorded three-centre failures

Earlier work in this programme recorded that every three-centre system attempted descended monotonically inward through its experimental bond, across nuclear charges, with and without a core, for odd and closed-shell systems alike. Measured by force at the observed geometry instead, CO_2 gives -7.2% and HCN -7.8%: ordinary bounded errors of the same size as the diatomics, not a collapse. Those scans were assembled from an energy relaxation that we find oscillates by $\sim 0.03\,E_h$ out to 3×10^5 steps, read through a stopping criterion that compares energies between wall-clock samples and cannot detect it. The recorded failures are therefore consistent with artefacts of the scan rather than properties of the model, and the systems should be re-measured before any mechanism is proposed for them.

17 Limitations

The bond errors reported here are not competitive with established electronic structure methods, and are not offered as such; they measure where this real-space construction currently stands. Several specific limitations bound the numbers.

The construction is not parameter-free. Besides the declared spacing it carries the five coefficients of the free-boundary update listed in Section 4, four of which have measured sensitivities below three points on H_2 and one of which (μ, the front speed) has never been laddered on a molecule. It also takes the valence charges Z_x from separate atomic work, and the kinetic occupancy cut-off at 0.9 is a declared choice worth 6.6 points of binding energy. None of these is fitted to the quantities reported here, but none is derived either.

All calculations use bare kernels, so the deepest, $Z = 4$, sits at $Zh = 0.48$, far above the resolution guideline. Mesh ladders are deliberately *not* used to justify the numbers, for the reason given in Section 4; the one ladder we do report, for H_2's bond, shows the refinement sequence passing through experiment rather than converging on it, which is a limitation of the construction and not of the sampling.

The convergence problem is characterised but not solved. We can now measure electronic non-convergence, attribute it principally to shallow kernels, and exclude three rival explanations; we cannot yet remove it. Configurations containing shallow kernels or threefold and bent geometries remain unusable, which includes water and ammonia—a serious limitation for any application to hydrogen-bonded systems.

Measured geometries, force constants and dissociation energies are taken from the standard compilations [7, 8] and are recorded as unverified in the source files. Dissociation energies carry the further ambiguity of Section 6.1: D_0 is measured and D_e adds a spectroscopic zero-point correction, the two differ by six points for H_2, and only D_e is the quantity a model without nuclear motion computes.

Finally, the construction pins the electron count of each domain, so charge transfer between unlike atoms is representable only through the shape of each domain and not through its occupation. For homonuclear systems this costs nothing; for a heteronuclear molecule it constrains precisely the asymmetry that defines it.

18 Conclusion

The simulator is defined by a declared grid rather than by a convergence limit: one spacing for every molecule, with the atomic references that enter the binding energy becoming constants of the model rather than per-run bookkeeping. That choice gives up the description of the construction as parameter-free, and we have listed what it actually carries. What it gains is that each number is answerable: with the grid fixed there is no remaining adjustment to defer a disagreement to.

Hydrogen, the one system whose atomic reference involves no pseudokernel, comes out as a single defect rather than three. Its depth is right to a few percent—$0.1666 \pm 0.0018\,E_h$, falling between the two experimental conventions, with the uncertainty measured over a factor of two in the initialisation rather than estimated. Its gradient is right, the volume force predicting the model's own minimum to 0.2% from one calculation at the observed geometry. Its curvature is 49% too soft, and its bond is 52% too long *because* of that, through $\Delta R = F/k_{\mathrm{model}}$. One quantity is wrong and the other errors follow from it, which is a more useful finding than three independent discrepancies would have been, and it identifies the curvature as what any improvement has to address.

Three things this work establishes that bear on how the earlier protocol should be read. The residual total force is a convergence diagnostic and not a figure of merit: an unbound H_2 scores as well on it as the best entries in the table. Bond errors converted with the measured force constant are lower bounds, too small by the stiffness ratio wherever the model's well is soft. And refining the mesh carries H_2's bond *through* experiment rather than towards it, so the construction's agreement with measurement lives at a finite spacing—which is the strongest argument for declaring the grid and, equally, the clearest statement of what declaring it costs.

What remains outstanding is specific rather than general: the front-speed coefficient has never been laddered on a molecule, the box dependence of the binding energy at fixed spacing is being measured, and the choice of interface condition—worth 29 to 58 points of binding energy—cannot be settled on any symmetric system, H_2 included, because symmetry makes the candidates coincide.

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[8] NIST Computational Chemistry Comparison and Benchmark Database, NIST Standard Reference Database 101.

[9] Full configuration interaction in a minimal (STO-6G) basis for equally spaced hydrogen chains H_N, generated for this work: equilibrium at $a = 1.8\,a_0$ with $E = -2.19038$ ($N = 4$) and -3.26674 ($N = 6$), each referred to its own atom $E(\mathrm{H}) = -0.471039$. The hydrogen chain is a standard benchmark for many-electron methods, having one electron per site and no pseudopotential, so no core parameter enters the comparison.

[10] Public code, the molecule simulator of Section 1, and an interactive gallery of all experiments in the present work: `https://claes542.github.io/RealMolecule/gallery.html` (source repository: `https://github.com/Claes542/RealMolecule`).

system	Z_x	force	ΔR	$\lvert\sum F\rvert/\lvert F\rvert_{\max}$	E_{bind} §
H_2	1,1	+0.1383	+30% ∗	14.7% †	**+0.1666**
N_2	3,3	−0.2505	−8.2%	3.7% †	**+0.2764** ¶
O_2	3,3	−0.2463	−14.3%	8.4% †	+0.0770
HCN	**1,4,3**	C≡N −7.8%, H–C +3.5%		**5.5**%	+0.6099
H_2O	3,1,1	—		126% ‡	+0.4307
H_3 lin.	1,1,1	+0.0420	—	34.9% †	—
H_4 chain	1,1,1,1	+0.0316	—	7.4–16% †	—
C_2H_2	1,4,4,1	outward	—	25% †	+0.3177
C_2N_2	3,4,4,3	N inward, C outward		3.2% †	+1.9819
NH_3	3,1,1,1	—		141% ‡	+0.5424
NH_3 planar	3,1,1,1	—		70% ‡	—
CH_4	4,1,1,1,1	0.036 outward	—	0.6% †	+0.4081

Table 2: Forces at measured geometries. The last column is the residual total force as a fraction of the largest single force; no verdict is attached, because what it *measures* depends on symmetry. †: every direction is symmetry-cancelled, so the entry reports whether the algorithm preserves symmetry and bounds nothing about convergence. ‡: cancelled except along the unique axis, so the residual is a convergence measurement in that direction. Unmarked rows (CO, HCN) have no symmetry, and their entries are genuine convergence measurements—note that these are among the largest in the table. Rows where no ΔR is given are those whose own checks exceed the signal. ∗: converted with the measured force constant, hence a lower bound; the model's own curvature gives +52% for H_2 (Section 6.3), and the same correction applies in unknown measure to every other ΔR here. §: only the H_2 entry (bold) is on the declared grid of (5) with the $w = 0.9$ cut-off; every other binding energy comes from an earlier set at the default valences in a box of $15\,a_0$ with $h = 0.1389$, $\varepsilon = 0.14$ and the uncorrected kinetic accounting. ¶: H_2 is the one row measured both ways, and it moves by $+0.0555\,E_h$, or 32 points of D_e, between them—6.6 of those points from the cut-off and the rest from the grid and softening. Every deficit in this column is smaller than that shift, so *the unbound entries are not results.* That has now been measured rather than suspected: re-run with molecule and reference on one grid (box 10, $h = 0.10$), **N_2 binds by** $+0.2764\,E_h$, from $E(\mathrm{N}) = -7.260914$ and $E(\mathrm{N_2}) = -14.798207$, with a symmetric state ($\Sigma F = 5.5\%$). The atomic reference shifts by only $0.008\,E_h$ between boxes 10 and 16, i.e. 6% of the binding energy, so the sign is secure. The earlier -0.0155 came from box 15 with $h = 0.1389$ and is withdrawn. Blank entries are rows whose valences or geometry were not in the earlier set; a row is deliberately left empty rather than filled from a run at different Z_x. There is no experimental column because the repository holds no verified table of atomisation energies, and H_2 is the only row with a reference we have checked. Configurations in which two different elements receive the same Z_x are degenerate—the model cannot distinguish them—and are omitted.